

PROMPTMINER-SAM2: PROMPT MINING AND UNCERTAINTY-GUIDED REFINEMENT FOR REMOTE SENSING INSTANCE SEGMENTATION

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Abstract—Overhead building instance segmentation is challenged by visually similar adjacent roofs, shadows, and narrow inter-building gaps. Two-stage detectors localize candidate instances, but a proposal box alone gives a promptable mask decoder little information about the target shape. We propose PromptMiner-SAM2, which maps proposal-conditioned features into SAM2’s prompt interface. An explicitly supervised coarse shape yields geometrically aligned point, box, and dense-mask prompts. To improve masks at ambiguous locations, an uncertainty-guided decoder-tail refiner updates low-confidence output locations. Experiments on the WHU dataset and NWPU VHR-10 benchmarks show that PromptMiner-SAM2 achieves 73.4% mask AP on the WHU dataset and 67.8% mask AP on NWPU VHR-10. Ablations further show that the complete tested configuration attains the highest mask AP and AP₇₅ among the ablated variants.

Index Terms—Instance segmentation, remote sensing, SAM2, prompt learning, building extraction

I. INTRODUCTION

Building instance segmentation from overhead imagery supports cadastral mapping, urban planning, and post-disaster assessment [18, 19]. The task is more demanding than semantic building extraction: a model must separate adjacent structures and recover each footprint despite small roofs, shadows, weak contrast, and narrow gaps [18, 19]. Two-stage instance segmenters address the *where* question by generating object proposals, but the proposal box alone supplies little information about which pixels inside that extent belong to the target instance [9, 11].

Promptable foundation models perform mask decoding conditioned on prompts. Segment Anything (SAM) accepts sparse point and box prompts as well as dense mask prompts [1]. Unlike SAM’s single-scale ViT encoder, SAM2 couples its native prompt interface with Hierarchical features [2], a natural fit for proposal-pyramid construction and prompt-conditioned mask decoding. A related effort has also scaled SAM-based segmentation to remote-sensing imagery [3]. A direct detector–SAM pipeline can use each pro-

posal as a box prompt. Yet box prompting leaves the decoder to resolve the target shape and to exclude nearby instances from the same region. This limitation is especially acute in overhead scenes with tightly packed buildings, where a proposal contains more instance-specific information than its rectangular extent alone. How to translate this proposal information into standard SAM2 prompts remains insufficiently explored.

Existing automatic-prompt approaches address related but different problems. RSPrompter learns category-related prompt embeddings from multi-scale features for remote-sensing instance segmentation [4]. AoP-SAM predicts prompt candidates and uses adaptive sampling and filtering to automate prompt selection efficiently [5]. PA-SAM inserts a trainable prompt adapter inside the SAM mask decoder to enrich sparse and dense *prompt features* with image details and mined hard points [6]. These methods learn prompt representations, select prompt candidates, or adapt decoder-side prompt features. SAMRefiner instead addresses universal mask refinement: given an initial coarse mask, it excavates distance-guided points, elastic boxes, and Gaussian-style mask prompts for a subsequent SAM refinement step [7]. PointRender likewise prioritizes uncertain locations for point-wise prediction, but does so through iterative subdivision [8]. These approaches do not specify how proposal-local shape information can be constructed and used as aligned SAM2 prompts within a two-stage instance-segmentation model for crowded overhead imagery.

We propose *PromptMiner-SAM2*, a two-stage model that couples a SAM2-based detector with SAM2 prompt-conditioned mask decoding. Hierarchical features form a detection pyramid for proposal discovery, while the SAM2 image embedding and high-resolution decoder features are used for final mask prediction. For each proposal, an explicitly supervised coarse shape yields foreground–background points and a dense-mask prompt; the proposal itself supplies the box prompt.

Our contributions are threefold:

- We adapt SAM2 to a two-stage instance-segmentation network by repurposing its hierarchical Hierarchical features as a PAFPN for RPN/RoI instance discovery, while re-

taining the SAM2 prompt encoder and mask decoder for final segmentation.

- We introduce CSPM, which converts a proposal-level coarse mask into aligned point, box, and dense-mask inputs for SAM2 decoding.
- We introduce an uncertainty-guided decoder-tail point refiner (UDPR) that directly interacts with SAM2 decoder logits and upscaled features to selectively calibrate low-confidence locations for localized mask refinement.

II. METHOD

A. Overall Architecture

PromptMiner-SAM2 couples a two-stage detector with SAM2’s prompt-to-mask interface. Hiera features support PAFPN/RPN/RoI proposal discovery, and CSPM transforms each proposal-local coarse shape into point, box, and dense prompts for adjacent buildings. Figure 1 presents the complete architecture and the data flow from proposal discovery to mask decoding.

Each proposal b_r anchors its multi-level RoIAlign feature X_r and all prompts in an image-aligned prompt coordinate frame. SAM2 decodes the image embedding with these proposal-specific prompts, and the resulting upscaled features enter UDPR. Points encode interior/exterior cues, the box constrains extent, and the dense prompt supplies a local shape prior.

B. Coarse-Shape Prompt Mining (CSPM)

For each proposal, the coarse branch maps its RoI feature X_r to a proposal-local shape logit map

$$C_r = f_{\text{coarse}}(X_r), \quad C_r \in \mathbb{R}^{1 \times 64 \times 64}. \quad (1)$$

Let $p_r = \sigma(C_r)$. PromptMiner derives a foreground support at probability 0.5 from p_r , while retaining the signed logits C_r for the dense prompt. It mines two positive and two negative points (2P2N) from this shared coarse shape. The first positive point is selected among confident foreground interior pixels ($p_r \geq 0.6$) by its foreground confidence weighted by its normalized distance to the support boundary; the second follows the same criterion while remaining spatially separated from the first. The first negative point comes from the one-pixel outer ring with low foreground probability ($p_r \leq 0.3$). The second is selected from background outside a dilation of the foreground support and is separated from the first positive and ring-negative points.

This construction assigns distinct geometric roles to the four points. The positive pair anchors the object interior without repeatedly sampling one local peak. In contrast, the ring negative is tied to the immediate exterior of the coarse

shape, whereas the second negative represents a less local background context. The 2P2N set records the coarse shape as sparse signed point prompts before they are transferred from the RoI frame to the image frame.

The resulting coordinates and positive/negative labels are mapped from the RoI frame to image coordinates. In parallel, the signed coarse logits are resized to the support of b_r and pasted onto the mask-input canvas:

$$S_r = \mathcal{M}(C_r, b_r), \quad Q_r = \text{Place}(C_r, b_r), \quad (2)$$

Here, $\mathcal{M}$ denotes 2P2N selection and coordinate mapping, and Place is the resize-and-paste operation. The two corners of b_r provide the box prompt. SAM2 jointly encodes the point, box, and dense-mask prompts, then decodes them with the image embedding in one pass:

$$Y_r^0 = \mathcal{D}(\mathcal{E}_I(I), \tilde{\mathcal{E}}_P(S_r, b_r, Q_r)), \quad (3)$$

where $\mathcal{E}_I$, $\tilde{\mathcal{E}}_P$, and $\mathcal{D}$ denote the SAM2 image encoder (including its decoder-side high-resolution features), complete SAM2 prompt-embedding pathway, and mask decoder, respectively. The pathway $\tilde{\mathcal{E}}_P$ includes the SAM2 composition of the dense-mask embedding. Q_r retains the signed rather than binarized coarse logits. Thus, the dense prompt preserves the spatial variation of the proposal-local shape, while the point and box prompts provide discrete geometric constraints to the same decoder.

C. Uncertainty-Guided Decoder-Tail Refinement (UDPR)

The same decoding pass also returns an upscaled decoder feature map U_r and the first output mask token t_r , in addition to the decoder logit map Y_r^0 . On the raster grid $\mathcal{G}_r$, UDPR visits the K_r locations with the smallest logit magnitudes:

$$K_r = \min(K, |\mathcal{G}_r|), \quad \Omega_r = \text{BottomK}(|Y_r^0|, K_r), \quad K = 64, \quad (4)$$

where BottomK returns the K_r indices in $\mathcal{G}_r$ with the smallest logit magnitudes. For each $i \in \Omega_r$, the aligned feature z_i is gathered from U_r and combined with the token, normalized location, local logit, and logit magnitude:

$$\phi_i = [z_i, t_r, Y_{r,i}^0, x_i, y_i, |Y_{r,i}^0|]. \quad (5)$$

The local feature describes the decoder output near i , the mask token conditions the update on the decoded instance, and the coordinate and logit terms specify the position and current decision state. A lightweight MLP g then predicts a bounded residual for each selected location:

$$\delta_i = 2 \tanh g(\phi_i), \quad i \in \Omega_r, \quad (6)$$

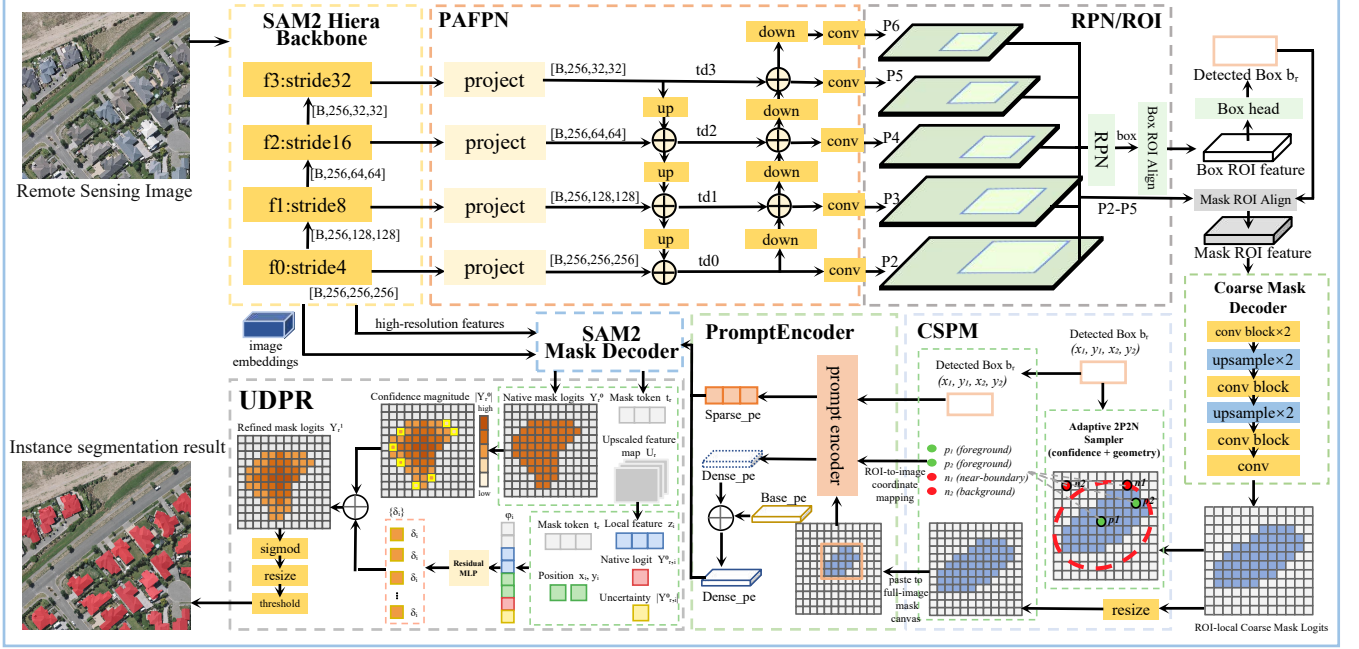


Fig. 1. Complete architecture of PromptMiner-SAM2. CSPM is the coarse-shape prompt mining module that converts each proposal-local coarse shape into point, box, and dense-mask prompts; UDPR is the uncertainty-guided decoder-tail refinement module that selectively refines uncertain decoder logits.

The refined logit map is formed by scattering these residuals back to their original positions:

$$Y_{r,i}^1 = \begin{cases} Y_{r,i}^0 + \delta_i, & i \in \Omega_r, \\ Y_{r,i}^0, & i \notin \Omega_r. \end{cases} \quad (7)$$

This selective update leaves all nonselected decoder logits unchanged and applies bounded residual corrections on the existing SAM2 decoder grid.

III. EXPERIMENTS

A. Training Details

Our PyTorch implementation uses distributed data parallelism on four NVIDIA RTX 3090 GPUs, with one image per GPU and two-step gradient accumulation (effective batch size eight). Both benchmark runs initialize from SAM2 Hiera-Base+ weights: the base Hiera parameters remain frozen and its LoRA adapters and the mask decoder are updated. We use AdamW with an initial learning rate of 5×10^{-4} and cosine annealing. Training uses cross-entropy, Smooth L1, binary cross-entropy, and Dice losses.

We select $K = 64$ by screening $K \in \{32, 64, 128\}$ under a common protocol; it achieves the highest overall bbox AP and mask AP in Table 1 and is used in the main experiments.

Table 1. Sensitivity of K on NWPU VHR-10 under a common development protocol. AP (%).

K	Bbox (%)			Mask (%)		
	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
32	69.38	91.69	79.03	64.94	92.91	70.01
64	70.35	92.23	81.65	66.65	94.15	71.97
128	70.13	92.42	81.32	65.97	93.71	72.22

B. Evaluation Protocol and Comparisons

We evaluate on the WHU dataset [20], a single-class building benchmark with 2,943, 627, and 2,220 training, validation, and test images, and NWPU VHR-10 [21], a ten-class benchmark. Since NWPU VHR-10 originally provides detection annotations only, we use the publicly released instance masks of Su *et al.* [22] and its 520/130 train/evaluation split. We use COCO-style bbox and mask AP, AP₅₀, and AP₇₅. To assess the proposed method, we compare it with 10 representative instance-segmentation methods. These methods fall into two broad groups: CNN-based methods, including two-stage, one-stage, and cascade designs; and methods that exploit global sequence modeling, including transformer-style mask prediction and foundation-model prompting.

Table 2 summarizes the quantitative results. PromptMiner-SAM2 obtains the highest mask AP among the compared methods on both benchmarks. On WHU, it reaches 73.4%

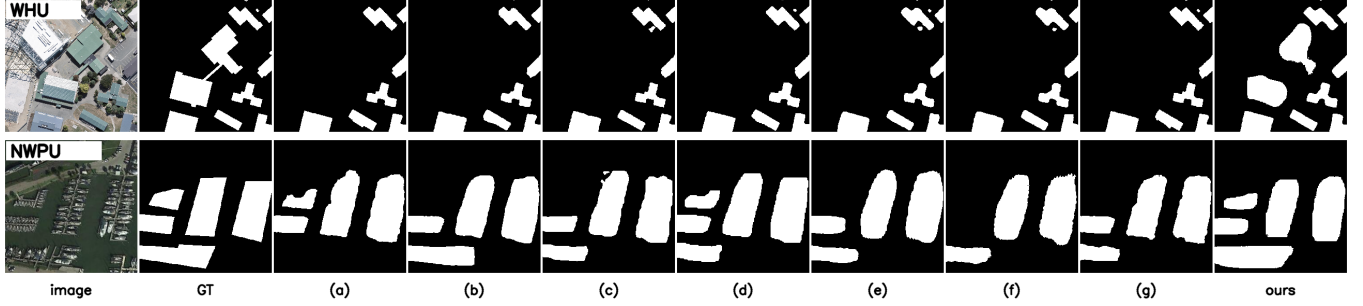


Fig. 2. Qualitative comparison on the WHU and NWPU VHR-10 datasets. From left to right, the columns show the input image, ground truth, predictions of (a) MaskDINO, (b) RSPrompter, (c) CondInst, (d) YOLO11, (e) HTC, (f) Mask R-CNN, (g) M2F, and PromptMiner-SAM2 (ours).

Table 2. Comparative analysis on the WHU and NWPU VHR-10 datasets. Each entry reports AP / AP₅₀ / AP₇₅ (%) for boxes or masks.

Type	Method	WHU						NWPU VHR-10					
		Bbox (%)			Mask (%)			Bbox (%)			Mask (%)		
		AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
<i>CNN-based</i>	Mask R-CNN [9]	70.9	89.9	80.9	68.2	90.0	80.0	65.4	90.2	78.9	63.0	90.1	71.2
	MS R-CNN [10]	68.8	89.4	80.0	67.7	89.6	79.5	67.8	94.3	83.8	64.7	93.6	80.8
	HTC [11]	72.7	90.6	82.5	69.2	90.7	81.6	58.9	86.9	67.7	51.3	84.6	53.3
	CondInst [12]	74.2	90.0	82.4	69.7	90.2	80.9	65.0	90.1	79.6	62.8	90.7	68.1
	YOLO11s-seg [15]	76.7	91.2	85.4	66.8	91.3	79.1	71.0	94.4	79.5	63.9	93.9	67.3
<i>Global sequence modeling</i>	Mask2Former [14]	61.4	82.7	70.7	64.5	84.1	75.6	60.3	77.9	66.8	61.5	84.9	67.8
	MaskDINO [16]	73.3	90.8	82.7	73.3	90.3	82.5	65.7	88.9	74.2	65.8	89.6	71.5
	RS4D-box [17]	64.5	87.6	74.7	64.1	87.8	75.2	57.0	82.5	65.4	54.3	80.1	57.6
	RSIISN [18]	74.1	90.0	84.4	72.1	90.1	83.9	66.7	91.5	78.4	62.9	89.7	67.1
	RSPrompter [4]	72.5	91.0	81.7	72.5	92.0	82.9	70.3	93.6	81.0	66.1	92.7	70.6
Ours	PromptMiner-SAM2	76.4	91.5	85.4	73.4	91.4	85.5	72.8	93.9	84.0	67.8	95.4	74.0

mask AP, and 85.5% mask AP₇₅. Its 76.4% bbox AP is 0.3 points below the highest bbox AP in this comparison. On NWPU VHR-10, it obtains the highest bbox AP (72.8%), mask AP (67.8%), mask AP₅₀ (95.4%), and mask AP₇₅ (74.0%) among the compared methods. The latter is 2.5 points above the next-highest mask AP₇₅.

Figure 2 juxtaposes the input image, ground truth, seven reference methods, and PromptMiner-SAM2 in a WHU building scene and a multi-class NWPU VHR-10 scene, providing visual context for the comparison.

C. Ablation Study

Table 3 compares prompt compositions and decoder-tail refinement under the same protocol. From P to $P+B$, mask AP/AP₇₅ increases from 63.0/67.4% to 65.3/71.7%. Adding M further raises mask AP to 65.9%, but reduces AP₇₅ from 71.7% to 71.5%. With UDPR, the full configuration attains 67.8% mask AP and 74.0% AP₇₅, the best mask results in this ablation.

Overall, enriching the prompt path raises mask AP from 63.0% to 65.9% before refinement, while UDPR further

improves both mask AP and AP₇₅. The complete configuration also achieves the strongest bbox AP and bbox AP₇₅ in Table 3, indicating that the full design improves the final instance-segmentation results under the shared setting.

Table 3. Prompt-path ablation on NWPU VHR-10. P , B , and M denote point, box, and dense-mask prompts.

Variant	Bbox (%)			Mask (%)		
	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
P	71.8	93.6	81.9	63.0	92.6	67.4
$P+B$	71.7	93.5	81.7	65.3	93.5	71.7
$P+B+M$	72.1	94.4	81.1	65.9	93.5	71.5
$P+B+M$ + UDPR	72.8	93.9	84.0	67.8	95.4	74.0

IV. CONCLUSION

PromptMiner-SAM2 integrates two-stage proposals with SAM2 through CSPM and UDPR. CSPM converts proposal-local shapes into aligned point, box, and dense-mask prompts, reducing ambiguity between adjacent build-ings. UDPR selectively refines low-confidence decoder logits. The resulting design performs well on both datasets.

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